

Sales Data Analysis Dashboard -Data Storytelling & Statistical Validation

Data Storytelling & Statistical Validation

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Task 4

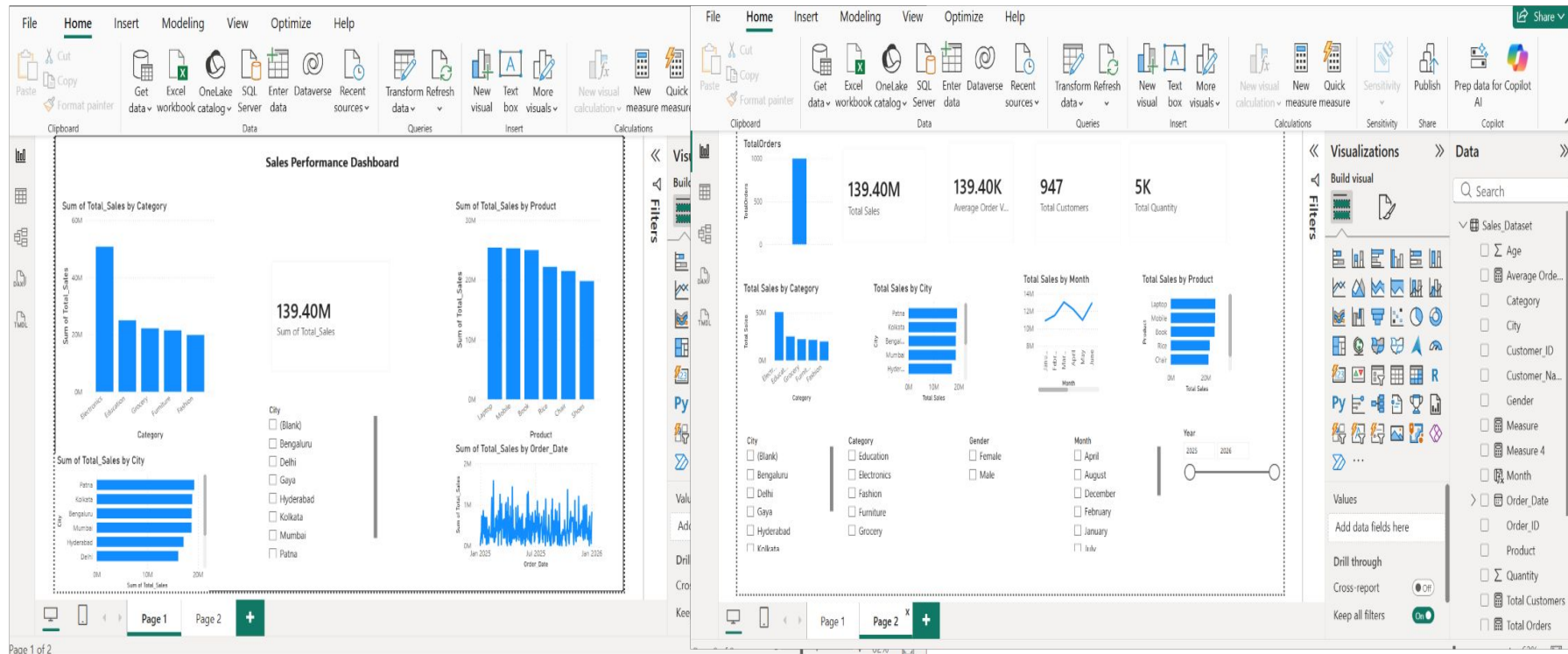
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Objective

- Analyze sales performance.
- Identify top-selling products.
- Compare sales across cities.
- Study monthly sales trends.
- Support data-driven business decision-making.

Interactive Power BI Dashboard



The dashboard provides an interactive view of sales performance across products, cities, categories, and months using KPIs and visualizations.

Key Findings

-  Total Sales: **139.40M**
-  Total Orders: **1000**
-  Total Customers: **947**
-  Total Quantity Sold: **5000**
-  Highest Sales Category: **Electronics**
-  Top-Selling Product: **Laptop**
-  Highest Sales City: **Patna**
-  Peak Sales Month: **March**

Data Story

The Power BI dashboard provides a comprehensive view of sales performance. Electronics generated the highest sales, with Laptop emerging as the best-selling product. Sales varied across cities and months, revealing seasonal demand patterns. These insights help improve inventory management, marketing strategies, and overall business decision-making.

Business Impact



Helps identify top-performing products and categories.



Improves inventory planning by analyzing sales trends.



Supports targeted marketing based on customer demand.



Assists management in comparing sales across different cities.



Enables data-driven decisions to increase sales and profitability.

Hypothesis Testing

Business Question

Does product category significantly affect total sales?

H_0 (Null Hypothesis)

Product category has no significant effect on total sales.

H_1 (Alternative Hypothesis)

Product category has a significant effect on total sales.

Observation

The dashboard indicates that Electronics generated higher sales than other categories.

Business Interpretation

The dashboard suggests a relationship between product category and sales. However, an actual statistical test (such as ANOVA or Chi-square, depending on the data) would be required to confirm statistical significance.

Business Recommendations

- Increase stock for high-demand products like **Laptop**.
- Continue promoting the **Electronics** category.
- Develop marketing strategies for lower-performing categories.
- Analyze customer preferences regularly to improve sales.
- Use monthly sales trends for better inventory and sales planning.
- Focus promotional campaigns during high-sales months.

Conclusion

This project demonstrates how Power BI can transform raw sales data into meaningful business insights. The interactive dashboard helps analyze sales trends, product performance, and customer behavior, enabling better business decisions and supporting future growth.

Thank You

Thank You!

Thank you for your time and attention.

Questions?